

Restriction Enzyme Analysis of a Highly Diverged Satellite DNA from *Drosophila nasutoides*

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Abstract. The satellite II DNA of *Drosophila nasutoides* is a highly diverged repetitive DNA, showing about 17% base changes between repeat units (Cordeiro-Stone and Lee, 1976). This DNA is cleaved by four different restriction enzymes to produce multimeric fragmentation patterns, indicating that their restriction sites are regularly arranged. Moreover, all four enzymes produce identical fragment lengths, the size of a monomer being 96 base pairs. Such multimeric patterns are expected for a diverged repetitive DNA, since many restriction sequences could have undergone changes during sequence divergence. Further restriction analyses of this DNA by double digestions and cloning reveal that there are three different sequences in satellite II DNA with respect to the presence and the arrangement of various restriction sites (Fig. 7). As an example, one sequence contains many EcoRI sites and fewer HinfI sites (20% of EcoRI sites), which are arranged regularly. These observations suggest that satellite II DNA of *D. nasutoides* might have evolved through different modes of sequence divergence.

I. Introduction

Many eukaryotic organisms contain satellite DNAs in their genome. Even if their biological functions are obscure at present, their presence in constitutive heterochromatin and the genetic importance of heterochromatin such as the variegated position effect (Spofford, 1976) make these DNAs worthwhile objects of investigation. There are, in general, two types of satellite DNAs with respect to their sequence homogeneity. Some satellite DNAs consist of highly repetitive sequences of identical repeat units. Well-known examples are those from *D. virilis* (Gall et al., 1973). On the other hand, there are considerable sequence divergences between repeat units in some other satellites. The mouse satellite DNA belongs to this class (Southern, 1975; Biro et al., 1975). Both types are found in *D. melanogaster* (Peacock et al., 1973; Brutlag et al., 1977) and in *D. nasutoides* (Cordeiro-Stone and Lee, 1976). The rapid divergence of highly repetitive DNAs is not unexpected, since these DNAs are not known to have

coding functions and, therefore, they may not be under stringent selection pressure.

The satellite II DNA of *D. nasutoides* is a highly diverged repetitive DNA in that, on the average, 17% base changes are observed between repeat units, the length of a repeat unit being about 103 base pairs (bp) according to our previous studies (Cordeiro-Stone and Lee, 1976). We wished to investigate further the sequence organization of this DNA using restriction endonucleases and cloning. Our results suggest that this satellite DNA might have undergone different modes of sequence divergence from the original sequence during evolution. A preliminary account on some restriction analysis of satellite II DNA has been reported earlier (Philippsen et al., 1977).

II. Materials and Methods

a) DNA Isolation. The satellite II DNA was purified from frozen adults as described previously (Cordeiro-Stone and Lee, 1976). Basically, the purification step involves 2 to 3 cycles of preparative CsCl gradient centrifugation followed by 1 to 2 cycles of actinomycin D-CsCl gradients. Whenever necessary, the purity of DNA was checked in the analytical ultracentrifuge. Thus, the final preparation of satellite II is pure in terms of its density.

Isolation of plasmid DNAs (pBR 322 and recombinant plasmids) was carried out by a laboratory protocol used at Dr. D.S. Hogness's Laboratory at Stanford (R.P. Lifton, personal communication). This procedure involves; gentle lysis of cells, removal of chromosomal DNA by centrifugation, ethidium bromide-CsCl gradient centrifugation, removal of ethidium bromide by a Dowex column, RNAase treatment and phenol extraction.

b) Digestion with Restriction Enzymes. All the restriction enzymes were purchased from Bethesda Research Laboratory and Miles Biochemicals. Incubation conditions were as suggested by the manufacturers. DNA was routinely digested for 2 to 3 h at 37°.

c) Agarose Gel Electrophoresis. Agarose gel electrophoresis was done in a horizontal slab gel apparatus with constant voltage (150–200 V). A Tris-borate buffer containing 0.05 M Tris, 0.05 M boric acid, 0.001 M EDTA and 0.5 µg/ml of ethidium bromide was used both as a gel buffer and an electrode buffer. 3% agarose gel was used for molecular weight ranges of 50 to 400 bp and 1.5% gel for 300 to 1,500 bp. Gels were illuminated with a short UV lamp or light table and pictures were taken with a Polaroid using Type 55 P/N films.

d) Cloning of Restriction Fragments of Satellite II. Restriction fragments produced by EcoRI, Sall and AluI were cloned into EcoRI, Sall and PvuII sites, respectively, of pBR322. All the cloning experiments were performed under the NIH Guidelines for Recombinant DNA Research.

(i) Cloning of EcoRI fragments. To the EcoRI digest of pBR322 (46 µg), 3 units of calf intestinal alkaline phosphatase (Boehringer Mannheim) were added and the mixture was incubated at room temperature for 1 h. This step removes 5'-phosphates of vector DNA and thus increases the cloning efficiency, especially when restriction sites on vector DNA are outside of drug resistance markers. DNA was phenol-extracted twice and dialyzed against 0.01 M Tris and 0.001 M EDTA, pH=7.5 (TE) to remove phenol.

Two µg of EcoRI digest of satellite II was mixed with 1 µg of pBR322 DNA treated as above. The mixture was heated at 63°–65° C for 3 min. A ligation mixture was added to give 10 mM Tris, 10 mM (NH₄)₂SO₄, 10 mM MgCl₂, 10 mM dithiothreitol, 0.2 mM ATP and 1 unit of T4 DNA ligase (Bethesda Research Laboratory). The ligation was carried out at 10° for 18 to 24 h.

SF 8, a derivative of C600, whose genotype is *rK⁻ mK⁻ RecBC⁻ lop11 lig⁺ galΔ str^r* (R.W. Davis, personal communication), was used as host for transformation. The transformation procedure used in our experiments was slightly modified from Mandel and Higa (1970) and Cohen et al.

(1972) (J.R. Walker, personal communication). Cells were grown in L-broth (1% Bactotryptone, 0.5% yeast extract and 0.5% NaCl) to A_{600} of 0.5. Cells were pelleted and resuspended in one half volume of 0.05 M CaCl_2 –50 $\mu\text{g/ml}$ thymidine (CaCl_2 -TdR). After 30 min in ice, cells were pelleted and resuspended in 1/15 volume of CaCl_2 -TdR. After 5 min or longer in ice, 0.2 ml of cell suspension was mixed with 10 μl of ligated DNA and the mixture was kept in ice for 1 h. After a heat treatment of 2 min at 42°, 2 ml of L-broth was added and the mixture was incubated at 37° for 1.5 to 2 h. Transformed cells were plated on nutrient agar plates containing 25 $\mu\text{g/ml}$ tetracycline (Sigma) and 500 $\mu\text{g/ml}$ ampicillin (Sigma). Each colony was purified further for its genetic purity. As will be shown in Results, about 90% of the transformants contained recombinant plasmids.

(ii) *Cloning of SalI fragments.* The same procedure as above was used for cloning SalI fragments of satellite II. Since the SalI site on pBR322 is within the tetracycline resistance gene, we used both alkaline phosphatase treated and untreated pBR322 as a cloning vector. Colonies of ampicillin resistant transformed cells were picked and purified according to their sensitivity to tetracycline.

(iii) *Cloning of AluI fragments.* AluI fragments of satellite II were cloned into the PvuII site of pBR322 by blunt end ligation (Sugino et al., 1977). The ligation mixture was the same as above except that it contained 5 times more T4 DNA ligase and 60 $\mu\text{g/ml}$ of T4 RNA ligase.

III. Results

1. Multimeric Fragmentation Patterns of Satellite II Produced by Several Restriction Endonucleases

Satellite II of *D. nasutooides* is a highly diverged repetitive DNA, containing on the average, 17% base-pair changes between repeat units. In order to understand the modes of DNA sequence divergence during evolution, we analyzed the presence and the arrangement of restriction endonuclease cleavage sites on this DNA.

Out of many enzymes examined so far, four, EcoRI, SalI, AluI and HinfI, cleave this DNA to produce multimeric fragments. The lengths of fragments were the same for all four enzymes with the size of a monomer being 96 ± 3 bp. A display of multimeric patterns produced by HinfI, SalI and AluI is shown in Fig. 1A–C. The fragment lengths were determined using a HaeIII digest of Φ XRF or AluI and HinfI digests of pBR322 as molecular weight markers. To make sure that the fragments lengths produced by different enzymes were identical, mixtures of separate digests were analyzed by agarose gels as shown in Figs. 1E, F and 2D. No more heterogeneity in lengths was seen, compared to single digests, confirming that all those enzymes produced multimeric fragments of 96 bp monomers.

These observations immediately indicate that restriction sites by a given enzyme are regularly spaced in satellite II DNA. Such multimeric patterns are not unexpected for a diverged repetitive DNA, for enzyme sensitive sites might have become altered due to sequence changes during divergence. Similar observations were made previously by others on different systems (in mouse, Southern, 1975; Hörz and Zachau, 1977; in human, Manuelidis, 1978; Beauchamp et al., 1979; in *D. melanogaster*, Manteuil et al., 1975; Brutlag et al., 1977).

We next addressed the question of how these four restriction sites are arranged in the satellite II DNA sequence. Our results described above can be obtained from either one of the following models: (1) Each repeat unit contains all four restriction sites, some of which have become altered due to divergence.

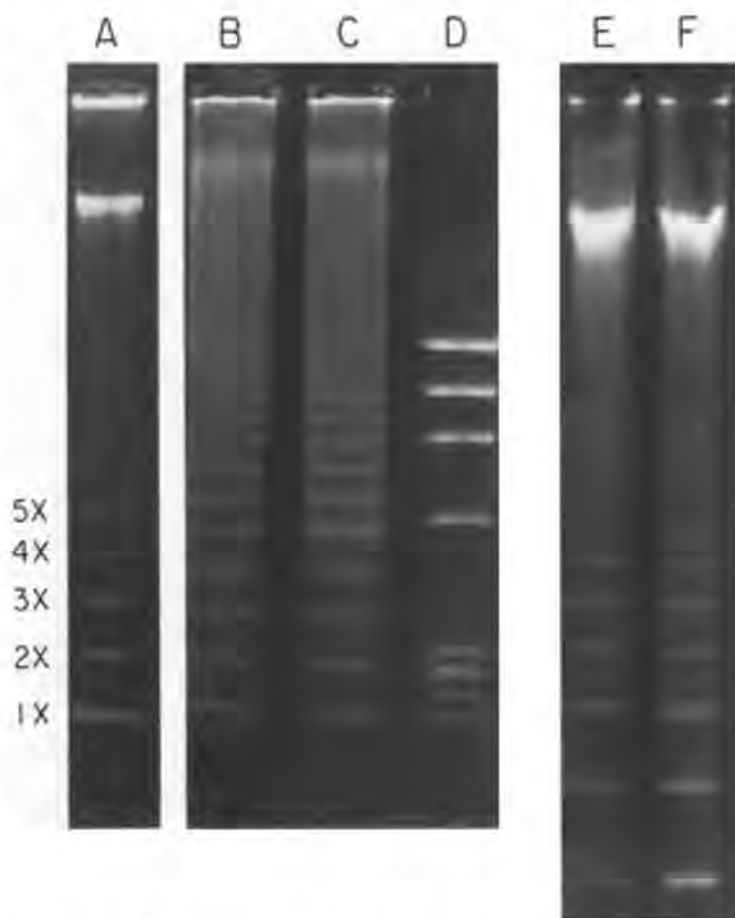


Fig. 1A-F. Gel electrophoretic patterns of satellite II digested with restriction enzymes. 1.5% gels were used. 1x, 2x, 3x, etc. indicate monomer (96 bp) dimer (192 bp), trimer (288 bp), etc. **A** *Hin*II digest; **B** *Sal*I; **C** *Alu*I; **D** *Hae* III digest of Φ XRF; **E** mixture of separate digests with *Sal*I and *Eco*RI; **F** mixture of separate digests with *Alu*I and *Hin*II

(2) There are four populations of sequences in satellite II, each of which contains only one type of restriction site. For example, a given sequence may have only *Eco*RI sites, some of which have been altered, to produce multimeric fragments upon digestion. (3) Some combinations of the above two extremes are also possible.

In order to resolve this issue, double digestion experiments were performed. If the model (1) above is correct, a digestion with two enzymes will produce intermediate fragments in addition to multimeric forms. If model (2) is correct, only multimeric fragments of 96 bp will result. The results of such double digestion experiments are shown in Figure 2. Figure 2A shows the fragmentation pattern produced by digesting satellite II with *Alu*I and *Hin*II. In addition to strong multimeric bands, faint intermediate bands were produced. The lengths

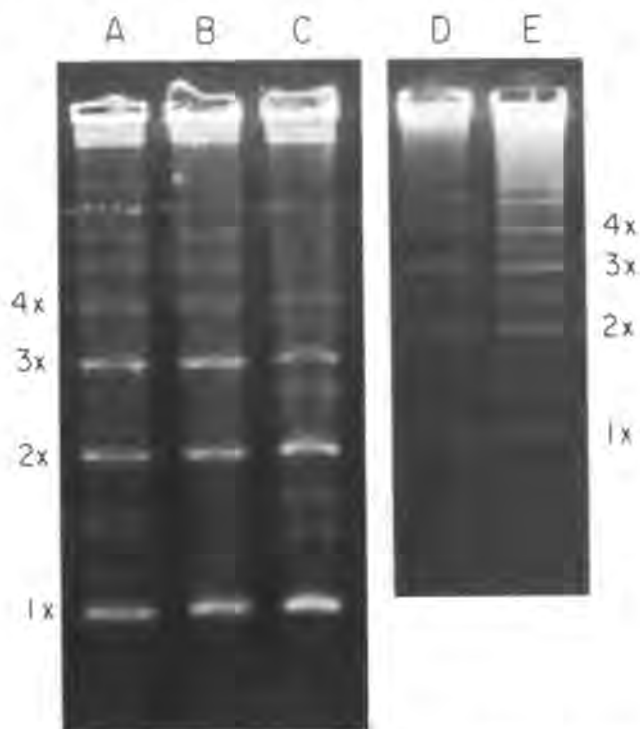


Fig. 2A-E. Gel electrophoretic patterns of satellite II digested with two restriction enzymes. DNA was incubated with the first enzyme for 1-2 h. The incubation mixture was adjusted for the second enzyme and incubated for 2 h more after the enzyme addition. 3% agarose gels were used. **A** Double digest with AluI and HinfI; **B** double digest with SalI and HinfI; **C** double digest with EcoRI and HinfI; **D** mixture of separate digests with AluI and SalI; **E** double digest with AluI and SalI.

of intermediate bands, for example, between the monomer and the dimer were 140 and 148 bp. Analysis of such a gel pattern reveals the arrangement of AluI and HinfI sites as shown in Figure 3c. Double digestion with SalI and HinfI does not reveal any detectable intermediate bands (Fig. 2B). Combinations of EcoRI and SalI and EcoRI and AluI also gave similar negative results. However, a double digest of EcoRI and HinfI displays reasonably strong intermediate bands (Fig. 2C). The lengths of intermediate bands, for example, between the monomer and the dimer are 134 and 154 bp, providing the arrangement of EcoRI and HinfI sites shown in Figure 3A. Figure 2E shows a double digest of AluI and SalI, and the resulting map is presented in Figure 3b. The results of all the double digestion experiments summarized in Figure 3 indicate that there are three different sequences with respect to their restriction site arrangements. It should be recalled that not all the restriction sites shown in this figure are sensitive to corresponding enzymes due to sequence changes.

This model of satellite II DNA, however, is incomplete for the following reasons: (i) Fainter intermediate bands suggest a predominance of one type of restriction site over the other, but we do not know which one. For example,

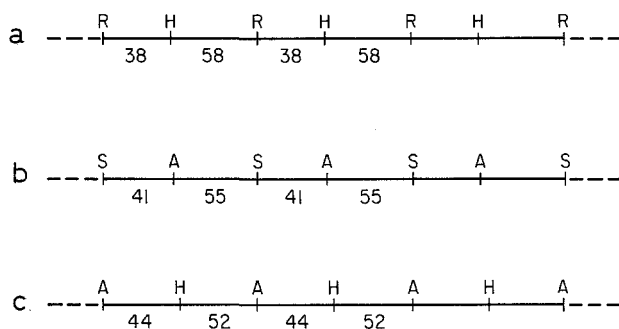


Fig. 3a-c. Possible restriction maps of satellite II based on double digestion experiments. *R*, *H*, *S* and *A* indicate EcoRI, HinfI, Sall and AluI sites, respectively. For convenience of presentation, all the restriction sites are shown, although some would have been altered by divergence and would not be cleaved. The numbers indicate lengths in bp between restriction sites. **a** Based on double digestion with EcoRI and HinfI; **b** based on double digestion with Sall and AluI; **c** based on double digestion with AluI and HinfI

in Fig. 3a we do not know whether EcoRI sites are more abundant than HinfI sites, or vice versa, in this sequence. All we know from the inspection of the gel (Fig. 2C) is that one of the enzymes cuts less frequently. (ii) Rare restriction sites for a third enzyme could exist within the models of Figure 3a, b or c, but they would not have been detected. (iii) If there is an additional restriction site very close to the first one, say 3 to 5 bp away, this site would have escaped detection. In fact, there is, often, a HinfI site 3 bp away from a Sall site; this point will be discussed in more detail below (p. 374).

2. Restriction Analysis of Cloned Satellite II DNA

In order to answer questions posed in the previous section and ultimately to sequence this DNA, we cloned restriction fragments into pBR322 as described in Materials and Methods. Purified plasmid DNAs were subject to various restriction enzymes and analyzed by gel electrophoresis as described below. Relevant restriction sites on pBR322 used in this study were obtained from Sutcliffe (1978a, 1978b).

(i) EcoRI Clones. Eighteen recombinant plasmids which contain EcoRI fragments were isolated and analyzed. EcoRI cleaved these plasmids to generate pBR322 and the cloned fragments. The lengths of the EcoRI fragments could be estimated directly; however, lengths were more conveniently determined using AluI. A 659 bp AluI fragment of pBR322 contains the EcoRI site and, therefore, AluI digests of recombinant plasmids will produce $(659 + N)$ bp fragments, *N* being the lengths of cloned fragments, provided that there is no AluI site in a clone. Figure 4A shows a new fragment of 950 bp, indicating insertion of a trimer size fragment. The new fragment in Figure 4B is 1,040 bp long indicating a tetramer insertion. There are no AluI sites in these two or any other EcoRI clones (see Table 1). Some examples of HinfI analysis of the EcoRI clones are shown in Figure 4. Figure 4C shows a HinfI digest of pBR322. A 1,631 bp HinfI fragment contains the EcoRI site and, therefore, HinfI digests

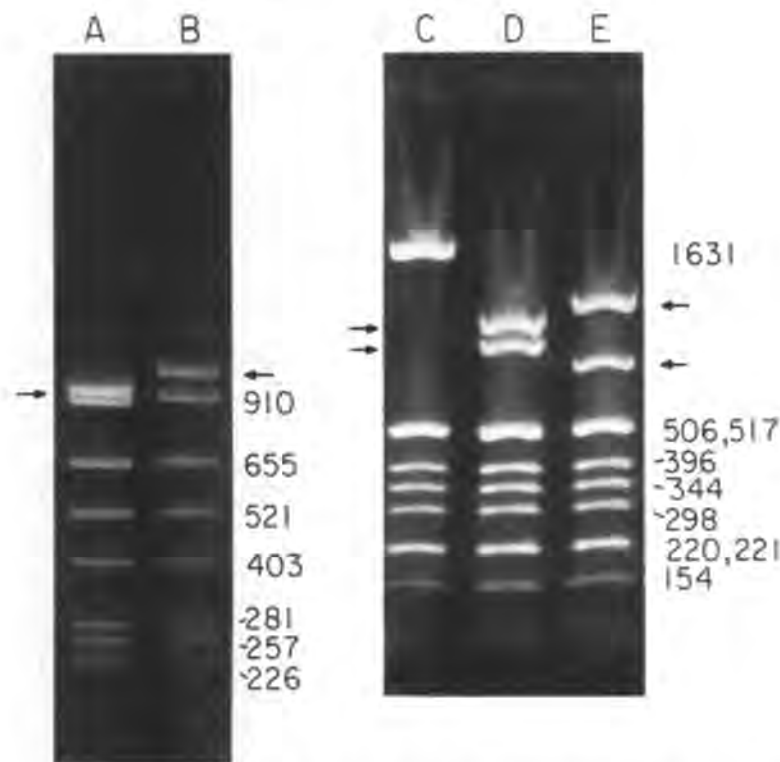


Fig. 4A-E. Restriction analysis of *EcoRI* fragments cloned into pBR322. Recombinant plasmids containing *EcoRI* fragments of satellite II were digested with various restriction enzymes and analyzed in 1.5% agarose gel electrophoresis. **A** *AluI* digest of a clone, R3, containing a trimer insert. Arrow indicates 950 bp fragment which resulted from a 288 bp trimer insertion into a 659 bp *AluI* fragment of pBR322. **B** *AluI* digest of a clone, R4, containing a tetramer insert. Arrow indicates 1,040 bp fragment which resulted from a 384 bp insertion into a 659 bp *AluI* fragment of pBR322. **C** *Hinfi* digest of pBR322. **D** *Hinfi* digest of a clone, R12, containing a trimer insert. Arrows indicate two new fragments of 895 and 1,025 bp produced by the presence of one *Hinfi* site in the clone. **E** *Hinfi* digest of a clone, R13, containing a tetramer insert. Arrows indicate two new fragments of 800 and 1,215 bp due to the presence of one internal *Hinfi* site in the clone

of recombinant plasmids should produce $(1,631 + N)$ bp fragments. However, if there is an internal *Hinfi* site in the clone, this DNA will be fragmented into two pieces. This is demonstrated in Figure 4D and E. Figure 4D shows a gel of a *Hinfi* digest of a recombinant plasmid containing a trimer size insertion. It can be easily seen that $(1,631 + N)$ bp fragment has been cleaved into two pieces of 895 and 1,025 bp due to an internal *Hinfi* site. More precise locations of *Hinfi* sites were determined by double digestion experiments with *EcoRI* and *Hinfi*. Restriction maps of two *EcoRI* clones are shown in Figure 6a and additional data are summarized in Table 1. Locations of *Hinfi* sites with respect to *EcoRI* sites agree well with the results previously obtained from double digestion experiments (Fig. 3a). Furthermore, there is approximately one *Hinfi* site for every five *EcoRI* sites on the sequence shown in Figure 3a (Table 1).

Table 1. Summary of restriction analysis of cloned satellite II

Enzymes used for cloning	Cloned fragment sizes	No. of clones analyzed	Occurrences of other restriction sites ^a
EcoRI	monomer (96 bp)	9	$\frac{\text{HinfI sites}}{\text{EcoRI sites}} = 20\%$; no SalI sites; no AluI sites
	dimer	2	
	trimer	4	
	tetramer	3	
SalI	monomer	5	$\frac{\text{HinfI sites}}{\text{SalI sites}} = 94\%$; $\frac{\text{AluI sites}}{\text{SalI sites}} = 24\%$; $\frac{\text{EcoRI sites}}{\text{SalI sites}} = 3\%$
	dimer	10	
	trimer	3	
	tetramer	5	
	pentamer	1	
	hexamer	1	
	heptamer	1	
	decamer	1	
	tetradecamer	1	
AluI	monomer	1	$\frac{\text{HinfI sites}}{\text{AluI sites}} = 22\%$; no EcoRI sites; no SalI sites
	dimer	3	
	trimer	2	
	2,276 bp	1	

^a The estimation of occurrences of other restriction sites was as follows: With EcoRI clones, for example, the total number of HinfI sensitive sites was divided by the total number of EcoRI sensitive sites in all the clones analyzed. Each cloned DNA fragment has at least two EcoRI sites, one on each end, by definition

(ii) *SalI Clones.* SalI clones were analyzed as described above. Some of the sample gels are shown in Figure 5. Figure 5A and B are gels of AluI and SalI digests of a clone, S505. SalI digestion revealed insertion of a tetramer size fragment. AluI digestion produced two new fragments of 280 and 763 bp. A 655 bp AluI fragment of pBR322 contains the SalI site and, therefore, AluI digest of recombinant plasmids should produce (655 + N) bp fragments, if there is no internal AluI site in the clone. Production of two new fragments (Fig. 5A) indicates an internal AluI site in this clone. The final restriction map of this clone with additional HinfI information (data not shown) is given in Figure 6b. Figure 5C and D are gels of AluI and SalI digests of a clone containing a dimer size insertion. Figure 5E shows a HinfI digest of a clone containing a hexamer insertion, showing some new bands (for example, 96, 128 and 298 bp) due to the presence of several internal HinfI sites in this clone. In fact, HinfI sites are as abundant as SalI sites in SalI clones (see Table 1), which was not obvious in double digestion experiments (Fig. 3). In many clones, HinfI sites are very near SalI sites, possibly arranged as:



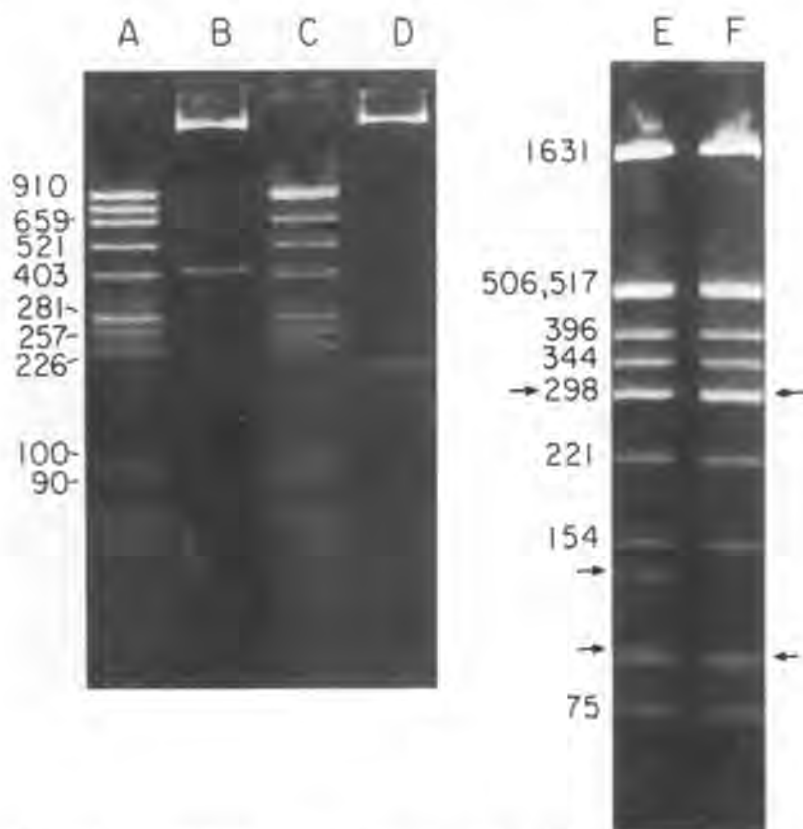


Fig. 5A-F. Restriction analysis of *Sall* fragments cloned into pBR322. Restriction digests were analyzed in 3% agarose gels to resolve smaller DNA fragments. **A** *AluI* digest of a clone, S505, containing a tetramer insert. There are two new fragments, one with a length of 763 bp and the other overlapping with the 281 bp fragment, which were produced due to the presence of an internal *AluI* site in this clone. **B** *Sall* digest of clone S505. **C** *AluI* digest of a clone, S509, containing a dimer insert. A new fragment of 850 bp was produced (655 bp+dimer length). **D** *Sall* digest of clone S509. **E** *HinFI* digest of a clone, S537, containing a hexamer insert. The appearance of several new bands, 298, 128, and 3 copies of 96 bp is due to the presence of many internal *HinFI* sites in this clone. **F** *HinFI* digest of a clone, S536, containing a heptamer insert. New fragments due to internal *HinFI* sites were 2 copies of 298 bp and 3 of 96 bp

The reason we did not see this in double digestion experiments described earlier is as follows: the frequency of sensitive *Sall* sites among all potential *Sall* sites, that is sensitive and altered (due to base changes), is about 30%. If we assume a similar frequency for *HinFI* (it may be higher due to less specificity of *HinFI* sequence), the chance of having two sensitive sites adjacent to each other as shown above is around 10%. The occurrence of 10% of 93 bp fragments among 96 bp fragments is difficult to detect in a gel system we used.

(iii) *AluI* Sites. Because of the inefficiency in blunt end ligation and tediousness in screening recombinant plasmids, only 7 clones were analyzed so far. Two

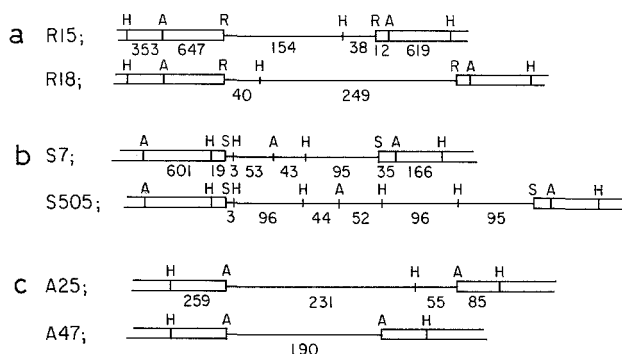


Fig. 6a-c. Samples of restriction maps of satellite II DNA cloned in pBR322. *H*, *A*, *R*, and *S* indicate *Hinf*I, *Alu*I, *Eco*RI and *Sall* sites. Numbers given are bp between restriction sites. Open thick bars indicate portions of pBR322. Positions of restriction sites are not always in scale. **a** *Eco*RI clones, **b** *Sall* clones, **c** *Alu*I clones

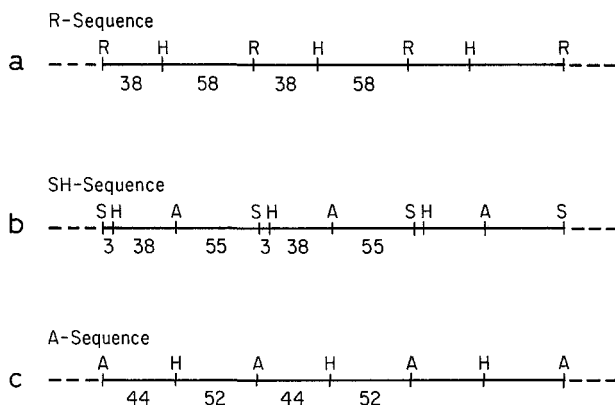


Fig. 7a-c. Restriction site arrangements in satellite II DNA. All the restriction sites are shown here regardless of whether they are cleavable or not. Relative frequencies of sensitive restriction sites are shown in Table 1

of these restriction maps are shown in Figure 6c and limited data on *Alu*I clones are summarized in Table 1. It is noted that there is approximately one *Hinf*I site for every five *Alu*I sites among *Alu*I clones.

3. Arrangements of Restriction Sites on Satellite II DNA

Combination of all the data described above results in the pattern of restriction site arrangements as shown in Figure 7. For convenience of presentation, all the restriction sites are shown regardless of whether they are sensitive or resistant (due to base changes). There are three different types of sequences in satellite II, R-sequence, SH-sequence and A-sequence. R-sequence contains many *Eco*RI and some *Hinf*I sites regularly arranged as shown. This sequence does not seem to contain either *Sall* or *Alu*I sites (Table 1). SH-sequence contains approximately equal numbers of *Sall* and *Hinf*I sites, often the distance between them being 3 bp. *Alu*I sites are between the *Sall* and *Hinf*I sites and are less frequent; *Eco*RI sites are rare (Table 1). A-sequence DNA has repeating *Alu*I sites and

contain 22% as many *Hinf*I sites. A comparison between SH-sequence and A-sequence predicts that there should be some *Sall* sites in A-sequence which was not observed. The reason probably is due to a small sample size of *Alu*I clones analyzed. The presence of such different sequences in the satellite II DNA suggests that there may have been different modes of sequence divergence of this DNA during evolution.

IV. Discussion

Multimeric cleavage patterns by restriction enzymes indicate that their restriction sites are regularly spaced. The occurrence of higher multimers could be due to sequence changes in restriction sites or to methylation or to both. The results of our cloning experiments eliminate the possibility of methylation as a cause for insensitive restriction sites. In order to support our assertion that random base changes on preexisting cleavage sequences cause multimeric fragmentation patterns, we have done a simple model study using a computer. Given a hypothetical genome consisting of 100 repeat units with each unit being 100 bp long and a 6 bp long cleavage sequence in it, we asked a computer to go through random base pair changes and cleave the resulting DNA. Figure 8 displays cleavage patterns at different extents of base changes. Our restriction patterns of *Eco*RI, *Sall* and *Alu*I (Fig. 1) resemble those obtained for 15–20% base changes (Fig. 8). This agrees very well with the 17% base change we reported earlier from our melting studies (Cordeiro-Stone and Lee, 1976). These studies suggest that the original repetitive genome contained restriction sites in each repeat unit but many of these sites have undergone sequence changes during divergence.

However, there are three different sequences in satellite II DNA with respect to the presence and the arrangement of different restriction sites (Fig. 7). This

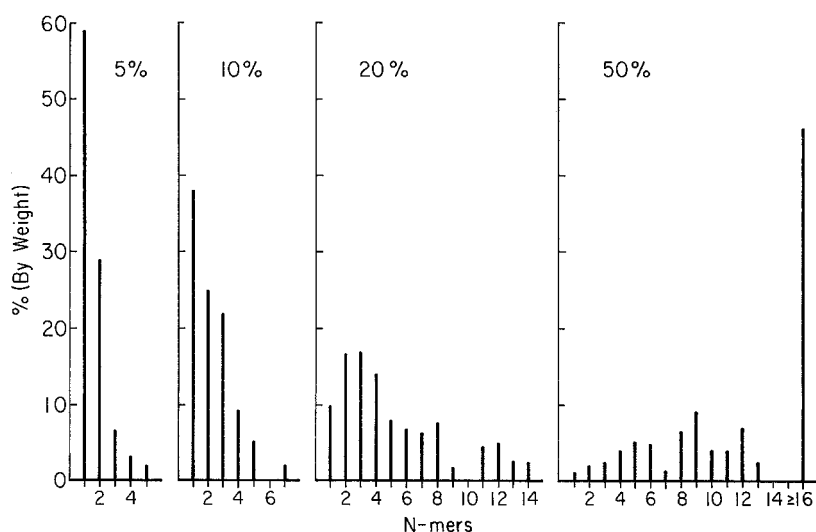


Fig. 8. Restriction patterns of a hypothetical repetitive genome undergoing various extents of random base changes generated by a computer (see text for details). The % in each panel indicates the extent of base changes

DNA is pure in terms of its density in CsCl gradients, and is resolved into two homogeneous bands in alkaline CsCl gradients (Cordeiro-Stone and Lee, 1976). Furthermore, the melting profile of the reassociated DNA shows a smooth transition. All of these observations indicate that the three different sequences mentioned above have similar base compositions, (G+T) distributions between complementary strands and extents of base changes. These characteristics and the fact that restriction sites are regularly spaced (identical for all four enzymes) suggest that these three different sequences might have evolved from a common sequence of 96 bp long. This unit must have undergone saltatory replication (Walker, 1971) or unequal crossover (Smith, 1976) To produce several copies, each of which later acquired or lost certain restriction sites. These sequences might have been amplified further and diverged during evolution to produce the modern sequence of this DNA. At present, we do not have any sequence information other than restriction sites on this DNA. Clearly, direct sequencing of this DNA will help in understanding the processes of its sequence divergence. This work is in progress.

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